

John Joseph Youkhana

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science in Electrical Engineering

Aug 2024 – May 2027

- **Relevant Coursework:** Control Systems (ECE 486), Power Electronics & Laboratory (ECE 464/469), Digital IC Design (ECE 482), Analog IC Design (ECE 483), Wireless Communication Systems (ECE 453), Power Circuits & Electromechanics (ECE 330), Digital Signal Processing (ECE 310)

Wilbur Wright College

Chicago, IL

Associate of Science in Electrical Engineering

Aug 2022 – May 2024

PROJECTS

Reaction Wheel Pendulum Control System

May 2026

- Designed and implemented an observer-based state-space controller for a reaction wheel pendulum, estimating unmeasured velocity states to achieve swing-up and inverted stabilization via DC motor torque.
- Derived nonlinear system model and conducted system identification to estimate plant parameters through experimental data analysis. Implemented real-time friction compensation in Simulink using PI control to compensate for coulomb and viscous friction effects.
- Designed and implemented an observer-based state-space controller to estimate unmeasured velocity states and stabilize the pendulum in the inverted position using reaction wheel torque.

Low-Dropout Regulator

May 2026

- Designed a CMOS LDO in Cadence 180nm technology, delivering a 1.2V output voltage across a 0.1–10mA load current range.
- Implemented a folded-cascode amplifier architecture with reference voltage and feedback as an error amplifier.
- Optimized sizing and values for the pass transistor and feedback resistance; implemented pole-splitting compensation via gate-drain capacitance across the pass MOSFET.
- Achieved DC load regulation of $< 14\mu\text{V}/\text{mA}$ and DC line regulation of $< 60\mu\text{V}/\text{V}$ for a load current of 1mA.
- Minimized quiescent current and output noise while maintaining a loop-gain phase margin of 66.2° to 87° across the 0.1mA–10mA output current range.

AC-DC Power Supply and Voltage Regulator

May 2026

- Designed and tested a regulated multi-stage AC-DC power supply; stepped down 120VAC to 25VAC via a transformer, converting to a stable 9.5V DC output with low ripple.
- Implemented the conversion stage using a full-bridge rectifier, capacitor filter, Zener voltage reference, op-amp gain stage, and BJT current boost stage.
- Designed a custom PCB in KiCad, optimizing trace widths for high-current paths through the rectifier and boost stages to manage thermal dissipation and minimize resistive losses.

EXPERIENCE

Electrical Team Member

Aug 2024 – Present

Illini Solar @ UIUC

Urbana, IL

- Designed PCB layouts in KiCad for a battery charge switch and horn oscillator, including component placement, routing, and DRC verification; assisted with board assembly of SMD and THT devices via soldering.
- Verified assembled boards through continuity and power checks to confirm functionality prior to integration.

Engineering Tutor

Aug 2023 – May 2024

City Colleges of Chicago at Wilbur Wright College

Chicago, IL

- Tutored engineering students in Calculus I-III, Physics I-II, Circuit Analysis, and C++ programming, providing one-on-one and small group support across core engineering coursework.

TECHNICAL SKILLS

Instrumentation: Power Meter, Digital Oscilloscope, Spectrum Analyzer, 3-Phase AC Power Supply, RF Signal Generator, Vector Network Analyzer (VNA)

Software & CAD: MATLAB, Simulink (Simscape, Stateflow), Cadence Virtuoso, Keysight ADS, LTspice, HSPICE, KiCad, AutoCAD

Programming: C/C++, Python (NumPy, SciPy), SystemVerilog, VHDL, Git